Notes pour la génération des fichiers lund.

Input:

- AHDC track
  - momentum range
- Electron
  - theta range
  - phi range
- Mining mass range W^2
- Delta phi range

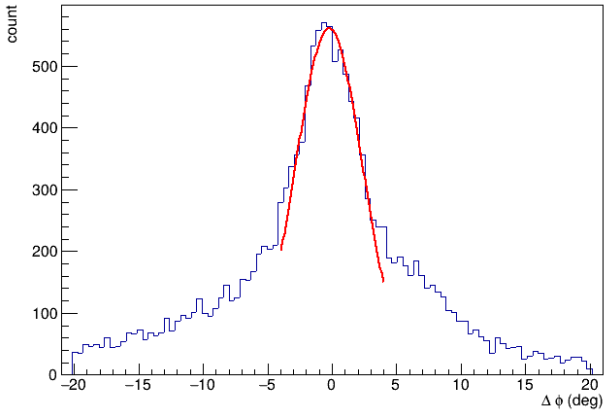
example: $p \in [200, 300]$

$\theta_e = 6.42 \pm 0.81$ deg

$\phi \in [0, 360]$ deg

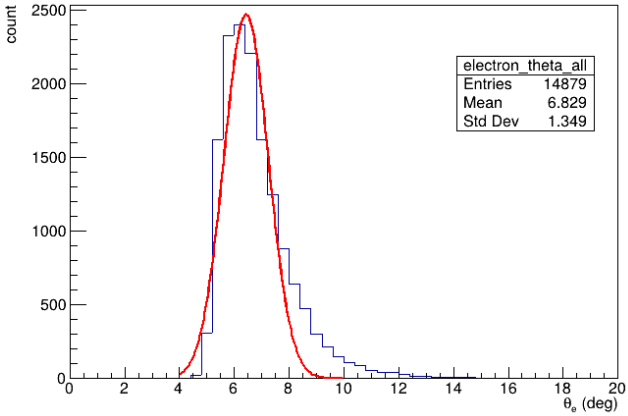
$W^2 = 3.64 \pm 0.11$ GeV^2

$\Delta\phi = -0.25 \pm 2.59$ deg



mean : -0.25294

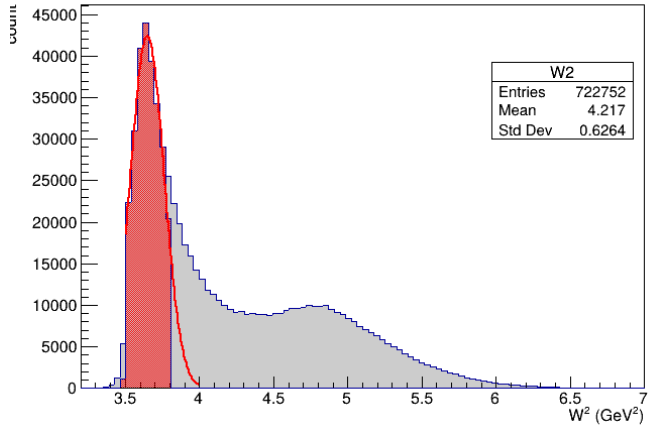
sigma : 2.59336



mean : 6.42737

sigma : 0.814156

W^2



mean : 3.64849

sigma : 0.113664

Equations:

$$Q^2 = 4EE'\sin^2\left(\frac{\theta_e}{2}\right)$$

$$W^2 = M^2 - Q^2 + 2M\nu \quad ; \; M: \text{target mass}$$

$$\nu = E - E'$$

Given θ_e, W^2, E:

$$W^2 = M^2 - 4EE'\sin^2\left(\frac{\theta_e}{2}\right) + 2M(E - E')$$

$$\Rightarrow W^2 = M^2 + 2ME - \left(4E\sin^2\left(\frac{\theta_e}{2}\right) + 2M\right)E'$$

$$\Rightarrow \boxed{E' = \frac{M^2 + 2ME - W^2}{4E\sin^2\left(\frac{\theta_e}{2}\right) + 2M}} \equiv \uparrow_e$$

Track kinematics:

$$\vec{\uparrow_e} \begin{pmatrix} -E'\sin(\theta_e)\cos(\varphi_e) \\ -E'\sin(\theta_e)\sin(\varphi_e) \\ E - E'\cos(\theta_e) \end{pmatrix}$$